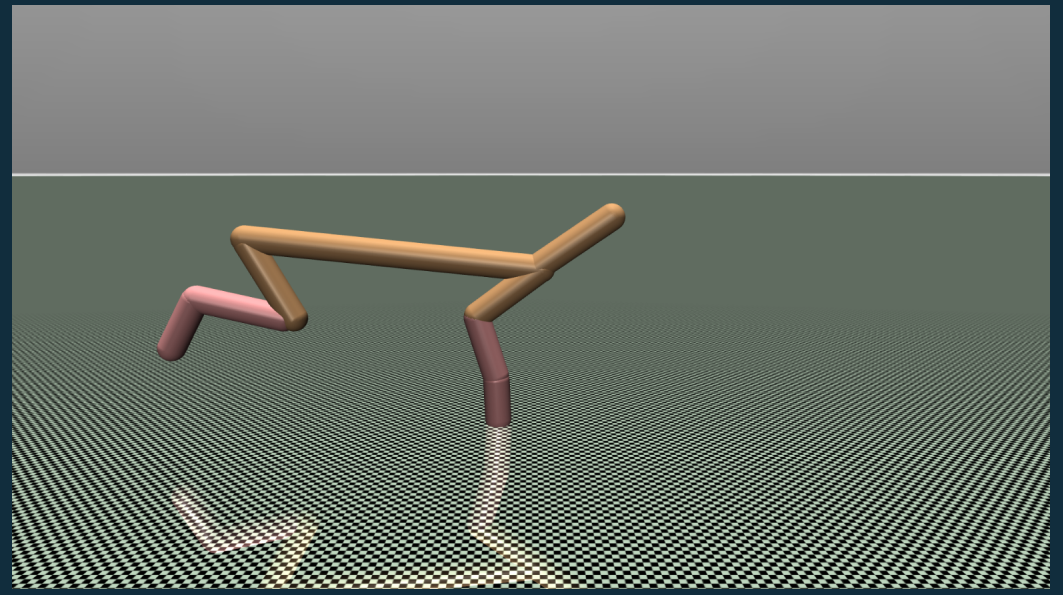
# Implicit Q-Learning

## HalfCheetah-v5



Offline reinforcement learning

Aleksandr Medvedev  Nikita Shankin  Leo Vesin
Ruslan Nasibullin  Telman Nuruzov

# How IQL learns from recorded data

## Offline, model-free actor-critic

**Recorded data**

Fixed transitions

(state, action, reward, next state)

**IQL learner**

Values and actor

**Policy**

6 bounded torques

**Simulator evaluation**

Complete policy episodes

## Better recorded actions receive more weight

# The three IQL updates

**01** **Value V**

Upper expectile

→ **Favor stronger actions**

expectile = 0.7

**02** **Twin critics**

Q1 and Q2

→ **reward + 0.99 × V(next state)**

Bootstrap unless truly terminated

**03** **Actor**

Weighted imitation

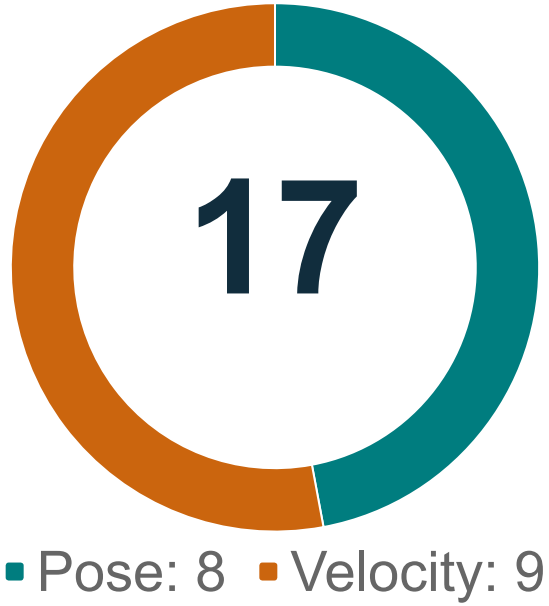
→ **min(exp(3 × advantage), 100)**

Advantage = min(target Q1, Q2) - V

Bounded policy: tanh-squashed Gaussian

# HalfCheetah observations and controls

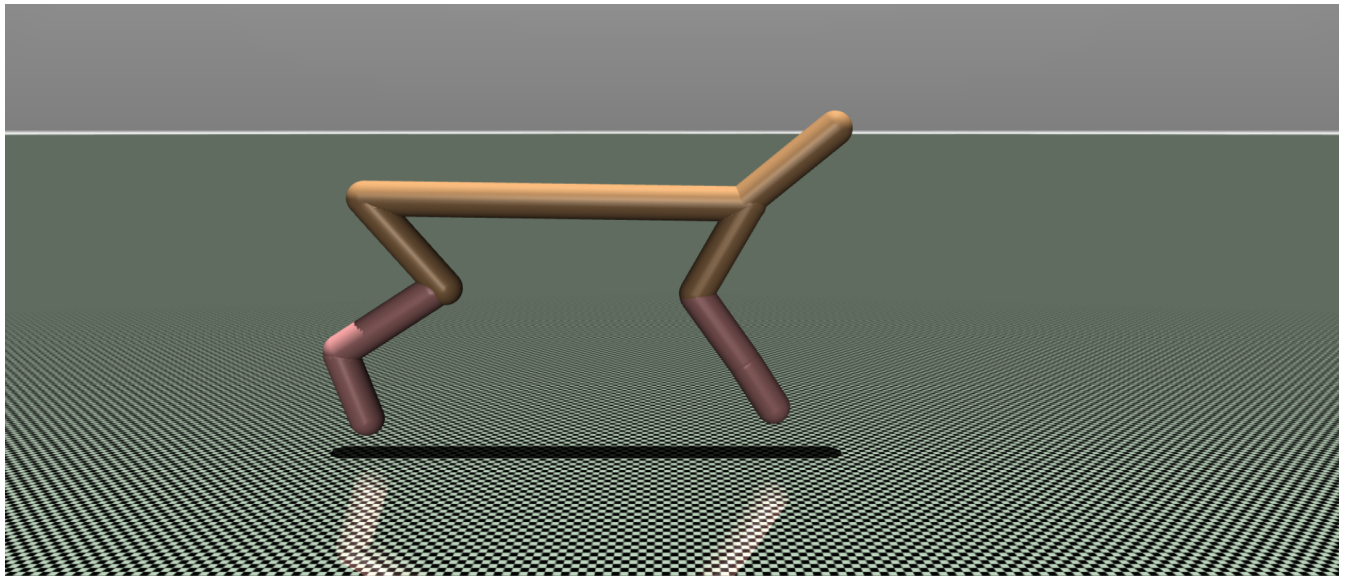
**17 observations**




Absolute x excluded

## 6 torques

**[-1, 1] per joint**



**2 legs × (thigh + shin + foot)**

# Reward and episode length

**Reward = forward velocity - 0.1 × sum(action²)**

Example: six action components of 0.5

| 5.00 | – | 0.15 | = | 4.85 |
|---|---|---|---|---|
| forward velocity | | control cost | | net reward |

**Goal: sustained, efficient forward motion**

---

**A fall does not end the episode**

**1,000 steps ≈ 50 seconds**

# Training and evaluation

**1M**
transitions

**3**
training seeds

**500k**
updates per seed

Minari medium: 1,000 episodes. Batch size: 256.

**Train**
Seeds 0, 1, 2

→

**Select**
10 eval episodes

→

**Final test**
New episode seeds

Raw return, random baseline, full videos. MuJoCo 3.2.3.

# Why HalfCheetah fits IQL

**Continuous torques** → **Continuous actor**

**Recorded locomotion** → **Offline learning**

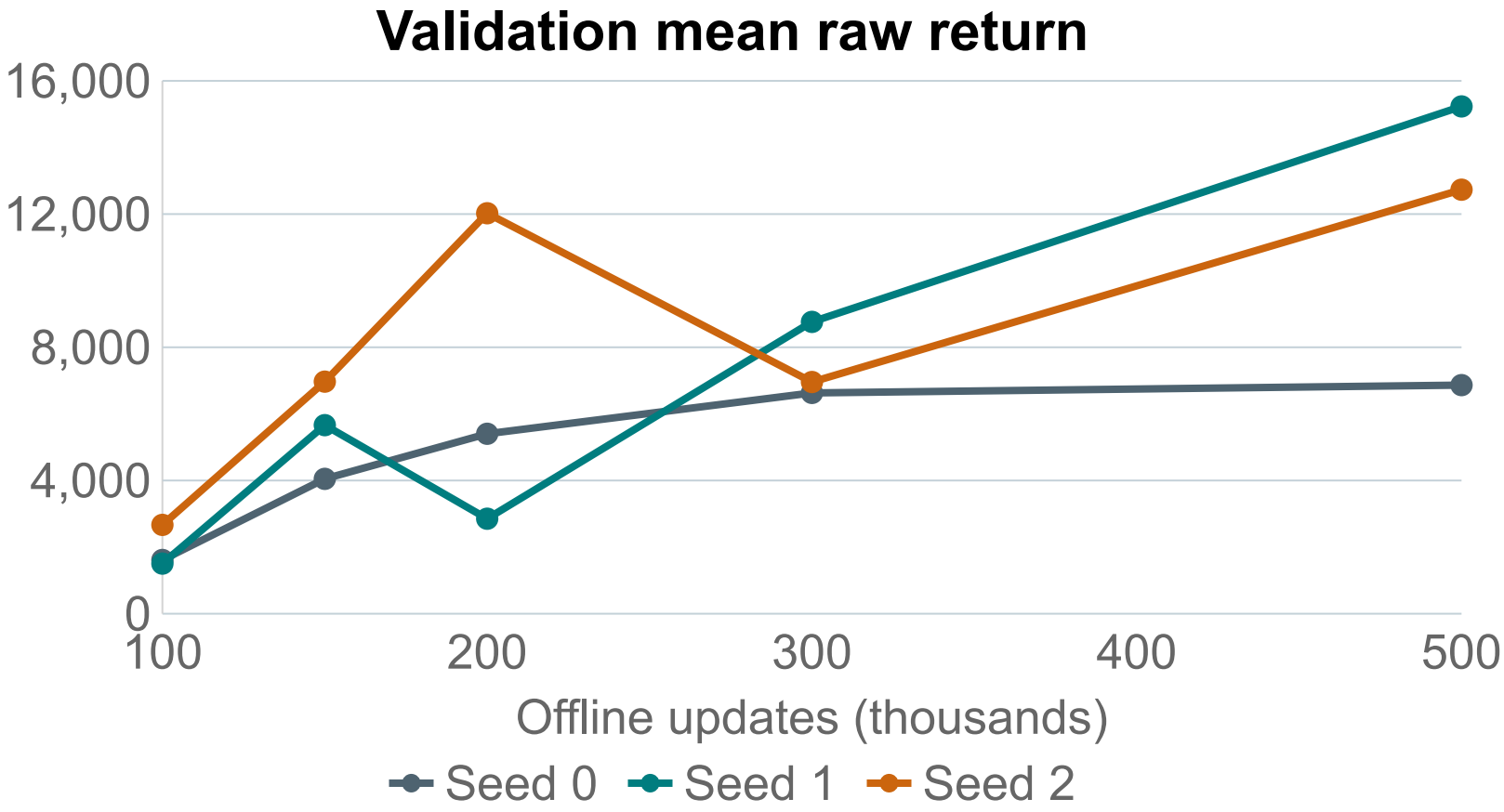
**Returns and visible gait** → **Measurable evaluation**

**Risks: data coverage, seed failures, physics differences**

Multiple seeds, complete episodes, pinned dependencies

Aleksandr: integration  Nikita: data  Leo: IQL
Ruslan: evaluation  Telman: notebook and presentation

# Learning curves and final test




**15,260**

final-test mean, 500k

vs 2,026 at 100k

**10/10 → 0/10**

episodes with long stationary tails

Means only, 10 episodes per point. Final test: seed 1, 10 new episodes.

**Other training seeds still fail**